P-Wave Detection Algorithm using phasor transform and template generation

1. Introduction to the Problem

The P-wave represents atrial depolarization, and it is vital in diagnosing cardiac conditions such as atrial fibrillation or atrioventricular blocks, and it's also very valuable when we are differentiating sinus arrhythmias and AFIB-like conditions. However, it is often considered difficult to identify the P-wave due to its nature. Here are some of them:

- P-waves have a low voltage, resulting in a low signal-to-noise ratio (SNR)
- P-waves have no exclusive time and frequency characteristics
- P-waves have high interpatient variability
- In the case of atrioventricular (AV) dissociations, P-waves do not respect normal time ordering of an ECG sequence and, thus, can be missing or redundant
- During tachycardia, P-waves can be hidden within the T-waves
- During atrial fibrillation (AFIB) and atrial flutter (AFL), P-waves are missing or replaced by so-called f-waves or F-waves, respectively
- In the case of ventricular ectopy, P-waves are typically absent.

You can find more about recent p wave detection approaches from this thesis

The goal of this algorithm is to accurately detect P-waves, including dissociated P-waves in AV II blocks, by combining template learning, phasor analysis, and RR interval logic. This is designed to operate without black-box learning and instead relies on a physiologically motivated approach based on heuristic rules, and a cardiac rule-based approach.

Approach so far:

1. First, I referred to the CEEMDAN-based ECG delineation method, and according to the resources found, seems like it can't be implemented in low-resource scenarios.
2. Then, I tried the template-based method suggested by this, and the major concern of the method is that it didn't contain a reliable approach to address pathological scenarios.
3. After that, I searched for the improved Pam Tompkins approach for the P-wave from this , and it seems to be not validated for enough pathological scenarios.
4. Following this, I focused on the phasor transform-based approach suggested by this. since it's a simpler and more recent approach, as mentioned in the literature. In this case, I initially tried applying the phasor transform all at once with the windowing approach for the whole signal, as mentioned in the paper. I somehow achieved permissible performance values greater than 80%. This was implemented as mentioned in the paper, but I couldn't achieve the exact performance levels as they mentioned. And couldn't find an obvious reason for the performance difference, and it need to be validated whether it's a human error or other form my side.
5. Then, along with that I developed a logic combining template generation of 1D arrays and the phasor transform with the windowing method, considering the simple approach. In this

method, I used the template-based method as suggested by this. As per the paper and the experimental results I observed that this method helps to enhance the P-peak from the noise since weight-based templates are mainly focused on the component that appears in more frequently in a common place in segments, thus reducing the effect of noise. The phasor transform further enhances the peaks, and then we can detect the peaks in the template and realign them to the relevant segment, obtaining the maximum of a small window around the signal. Instead of Shannon entropy and area under the R-peaks-based methods for AFIB and PVC detection, I directly used the annotated signals from the device for this since the final output of this would be run upon that.

6. Finally, the performance values were between 85-90% for all 4 datasets, and I fine-tuned the rules considering all 4 datasets. The final values were as below:

Dataset	PPV (%)	Sensitivity (%)
MIT-BIH (without 207)	**96.1**	**95.4**
BUT PDB	**89.6**	**89.1**
LUDB*	**90.4**	**97.5**
QTDB	**90.4**	**93.2**

2. Algorithm Description

Step 1: Load ECG and Annotations

Function: load_ann()

This step handles the data loading from an assigned path. The function loads .h5py formatted ECG signal files and/or their corresponding annotations for R-peaks and beat types.

- Loads the raw ECG signal and annotations.
- Extracts beat types:
  - Normal (code 78)
  - PVC (code 86)
  - AFIB (code 7 in column 2)
- Returns parsed signal and annotation indices for further use.

Step 2: Preprocessing

Function: filter_ecg()

ECG is preprocessed to remove slow-varying baseline noise and high-frequency artifacts. In this step, we use two filtering methods to achieve this. The baseline drift can be removed using techniques like high-pass filters, empirical mode decomposition, wavelet transforms, and other mathematical

morphologies. However, high-pass filters can alter the ECG morphology, and some of the other methods may require increased computational power. Therefore, we use a two-stage median filtering approach to reduce baseline drift, followed by Savitzky-Golay filtering, which preserves morphology while smoothing the signal.

1. **Apply median filter with kernel_size=fs-1 and 2*fs-1:**
    - The window sizes are set to the sampling frequency (fs) and twice the sampling frequency (2*fs). The output of the first stage is passed through the second stage, and the final output is subtracted from the original signal.
2. **Use Savitzky-Golay with window_length=30, polyorder=4:**
    - The output is then filtered with a Savitzky-Golay filter, which preserves low-frequency features like the P and T wave shapes while reducing noise. This least-squares smoothing filter utilizes polynomial fitting over the input data samples and estimates the resulting polynomial at a single data point within the window.

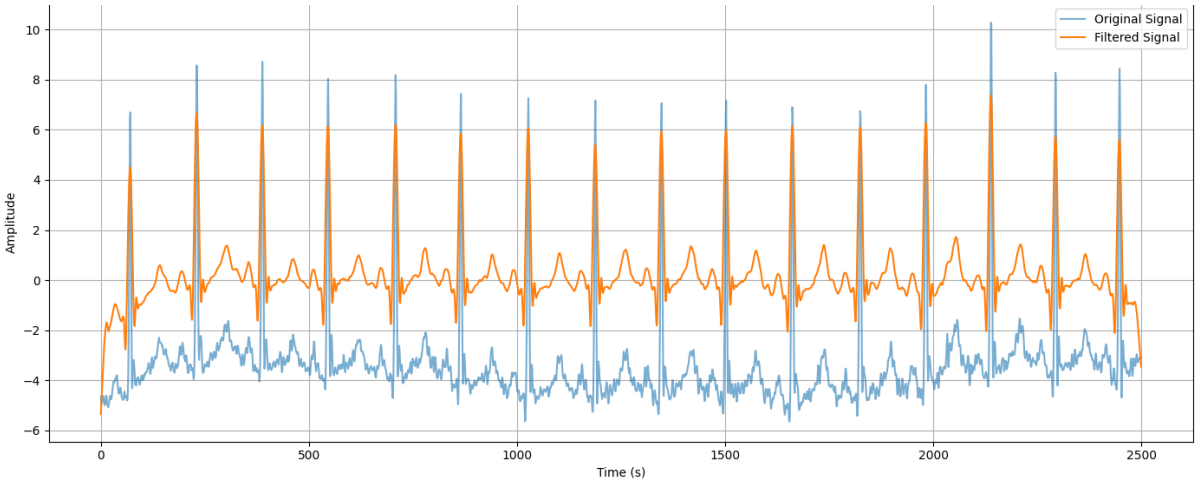


Figure 01: Raw signal (blue) and filtered signal (orange)

Step 3: Compute RR Intervals

Function: compute_rr_intervals()
RR intervals help define temporal windows for segmenting P- and T-waves. This calculates the duration between successive R-peaks.

- RR[i] = (R[i+1] - R[i]) / fs, Input: array of indices, output: each RR interval in seconds

Step 4: Pathology Filtering

Function: pathology_check()

Atrial fibrillation (AFIB) is characterized by irregular atrial activation, leading to the absence of clear P-waves in the ECG. Instead, fibrillatory waves appear there, which can easily confuse standard P-wave detection algorithms, resulting in numerous false positives.

According to clinical knowledge, if a current beat is classified as a PVC, there is typically no associated P-wave in the preceding RR interval. Therefore, accurate recognition of PVCs helps define segments where P-wave detection is unnecessary.

In the main method referenced, they used the area under the R-peaks to detect PVCs and Shannon entropy of RR intervals to detect AFIB. However, it haven't been included here, as highly verified alternative methods are already implemented in the device and can be used for this purpose. Furthermore, additional AFL detection logic and a method to reduce false positives in AFIB detection (to differentiate between AFIB and sinus arrhythmias) should be implemented.

Following the PVC and AFIB detection stages, the algorithm performs a pathology filtering. In this stage, signal segments identified as either PVC or AFIB are excluded from further P-wave analysis. If a beat is flagged by either of the above methods, the algorithm terminates processing for that interval and advances to the next beat.

Output: mark R-peaks with AFIB, PVC, or RR ≥ 3s as invalid for template generation.

Step 5: Segment Extraction

Function: extract_segments()

Segments within each valid R-R interval are extracted. Both P-wave and T-wave regions are defined empirically using a hybrid of literature-based and trial-and-error methods, validated across four datasets. Before segment extraction, each beat is checked using the valid_beats list. Only beats where valid_beats[i+1] is not None are processed. This ensures that invalid or noisy beats, such as AFIB or PVC, are excluded from P-wave and T-wave segmentation logic.

T-Wave Interval Estimation
For each valid beat (RR interval i), the T-wave region is defined relative to the current R-peak.

P-Wave Interval Estimation Based on RR Duration
This estimation is based on several logics, including the following cases:

- Short RR Interval: Typically occurs in tachycardia; P-waves are more left-shifted.
- Moderate RR Interval: P-waves move slightly rightward, with a heuristic offset added.
- Normal and Long RR Interval: Further adaptation based on signal polarity at the next R-peak.

Segment Quality Control

- If (end_p−start_p)<20 samples → the segment is skipped (replaced with an empty list).
- If RR interval > 3s → the entire beat is skipped (extreme bradycardia or invalid signal).

This initially used a very simple window definition, and accuracy was slightly lower. Therefore, I used a trial-and-error, cardiac knowledge-based method to define the windows more accurately.

Step 6: Template Generation

Function: generate_template()
Segments are grouped using Pearson correlation. This unsupervised step forms representative waveform clusters.

- Correlation > 0.8 – 0.9 to match an existing template
- Segments are resampled or interpolated before comparison to normalize length and the minimum length is 50 samples if the segment is less than that it is interpolated into 50 samples

This process helps to maintain the morphologies and noise conditions. It involves clustering signal segments based on shape similarity to create representative waveform templates for peak detection. Pearson correlation is used to group morphologically similar segments. In general, the similarity between two signals can be easily compared using the Square Sum Difference. However, a large error appears even when the signal shapes are similar if the baseline or amplitude scale changes, despite the signal being filtered. The proposed method applies Pearson similarity, which is robust to changes in baseline and amplitude scale.

First, the initial valid segment is stored as the starting template. New incoming segments are then compared to existing templates. If the correlation with any existing template exceeds the threshold, the segment is merged into that template using a weighted averaging method, as follows:

The weighted mean of the ith sample of the kth template CS (i, k) with the input signal S is calculated using the counter CC(k) as a weight, as shown in

$$C_S(i,k) = \frac{C_C(k) \times C_S(i,k) + 1 \times S(i)}{C_C(k) + 1}$$

The weighted mean, as opposed to the average, can reduce signal distortion even in the presence of an irregular beat. Furthermore, the counter serves as a weight, and there is no need for sorting. Separate template generation for each p and t segments were chosen because when we consider the whole segment, sometimes times p peak seems to merge into the baseline and then identify it as a false negative.

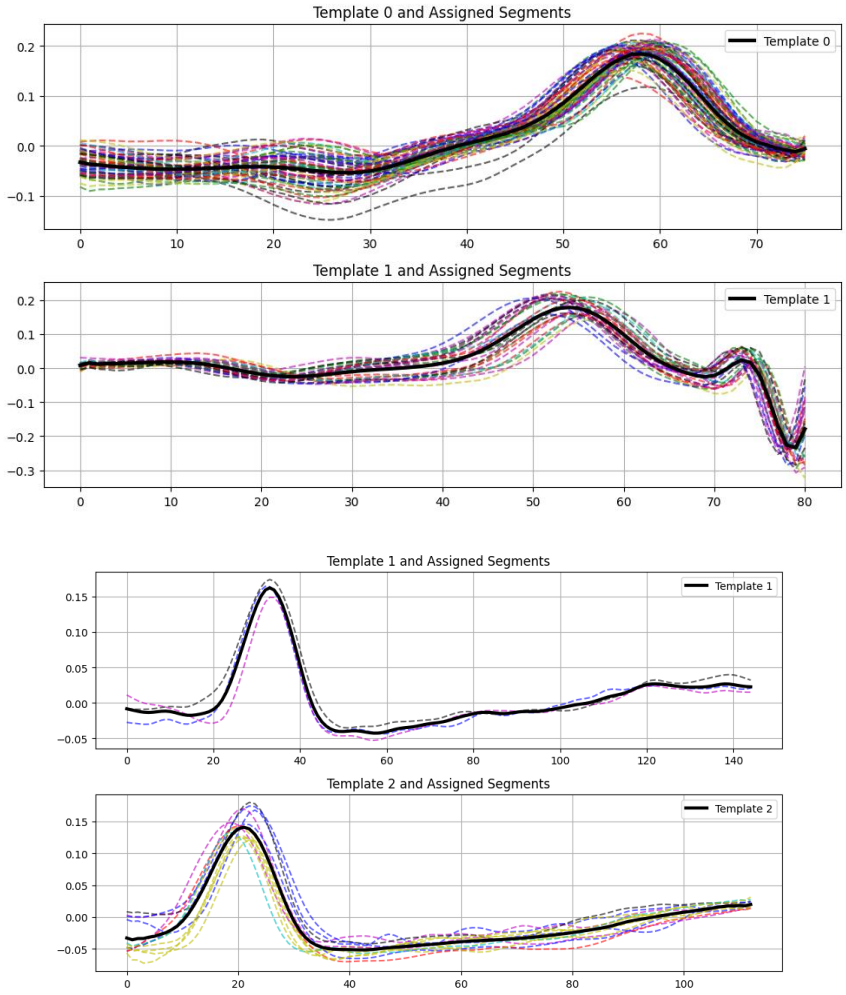


Figure 02: examples for template generation

Step 7: Drop Low-Weight Templates

Function: drop_low_weight_templates()
Underrepresented templates are filtered. Templates with too few segments or insufficient total contribution are discarded.

- Templates with <10 segments, till <5% of total beats are dropped or singleton templates are dropped only if the number of segments is more than 50.

Step 8: Peak Detection in Templates

Function: detect_peaks_in_templates() and find_best_peak()

This method uses phasor transformation and prominence*width-based scoring to detect the relevant peak from a template.

Each ECG segment or template is analyzed individually. For every template:

- The signal is phasor-transformed to emphasize frequency and phase dynamics.
- Both the original and inverted signals are considered to capture peaks with varying polarities.
- Detected peaks are scored based on amplitude and width.
- The most representative peak is selected through a heuristic approach.

In 2010, Martinéz et al. published an original paper on the automatic delineation of ECG fiducial points using phasor transform. PT enhances variations of the signals, thereby making the detection of the components easier. The PT transforms each sample of the original ECG signal, x(n), into a phasor signal, y(n), where n is discrete time. The real part of the signal is a constant value RV, which is set by the user, and the imaginary part of the phasor signal is the original value of the actual sample x(n). The phasor y(n)can be defined for each sample as follows:

$$y(n) = RV + jx(n)$$

From the phasor, magnitude M(n) and phase φ(n) are calculated according to below equations.

$$M(n) = \sqrt{R_V^2 + x(n)^2}$$

$$\phi(n) = tan^{-1}\left(\frac{x(n)}{R_V}\right)$$

Phase φ(n) is mainly used for P wave detection. This signal is normalized to the interval <0, π/2>. The efficiency of the enhancement of the ECG components can be controlled by setting R V. The value of R V is always within the interval <0, 1>. If the R V is low, phase differences are higher, and thus, changes in signal are highlighted.

The find_best_peak function then identifies peak candidates in both the positive and negative phasor-transformed signals:

- This is used to detect candidates:
  - Positive component with low prominence threshold (0.05)
  - Negative component with higher prominence (0.2) in PT signals.

Then, each peak candidate is assigned a score:

score =relative amplitude of the peak within a small window* peak_width

Where the relative amplitude is computed locally in a ±10 sample window in a shifted signal. Then the highest-scoring peak is identified as the peak position.

The best peak is chosen based on the following cases:

- **Only positive peaks found:** Select the one with the highest score.

- **Only negative peaks found:** Same logic applies.

- **Both sets found:**
    - Compare the best positive and best negative peak scores.
    - Select the negative peak only if its amplitude is more than 3× the positive one (this is heuristically decided since negative p peaks are rare and if exist, they have a considerable amplitude.).

- **If this is for dissociated peaks case:**
    - Only consider positive phasor peaks.
    - If none exist, return none.

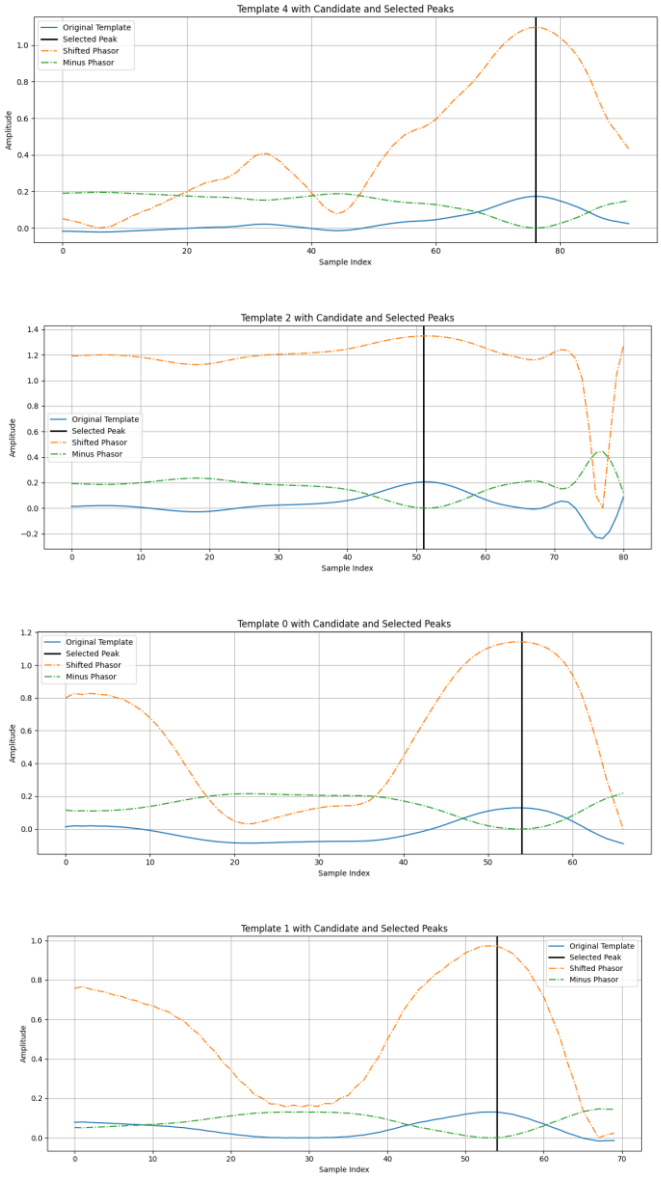


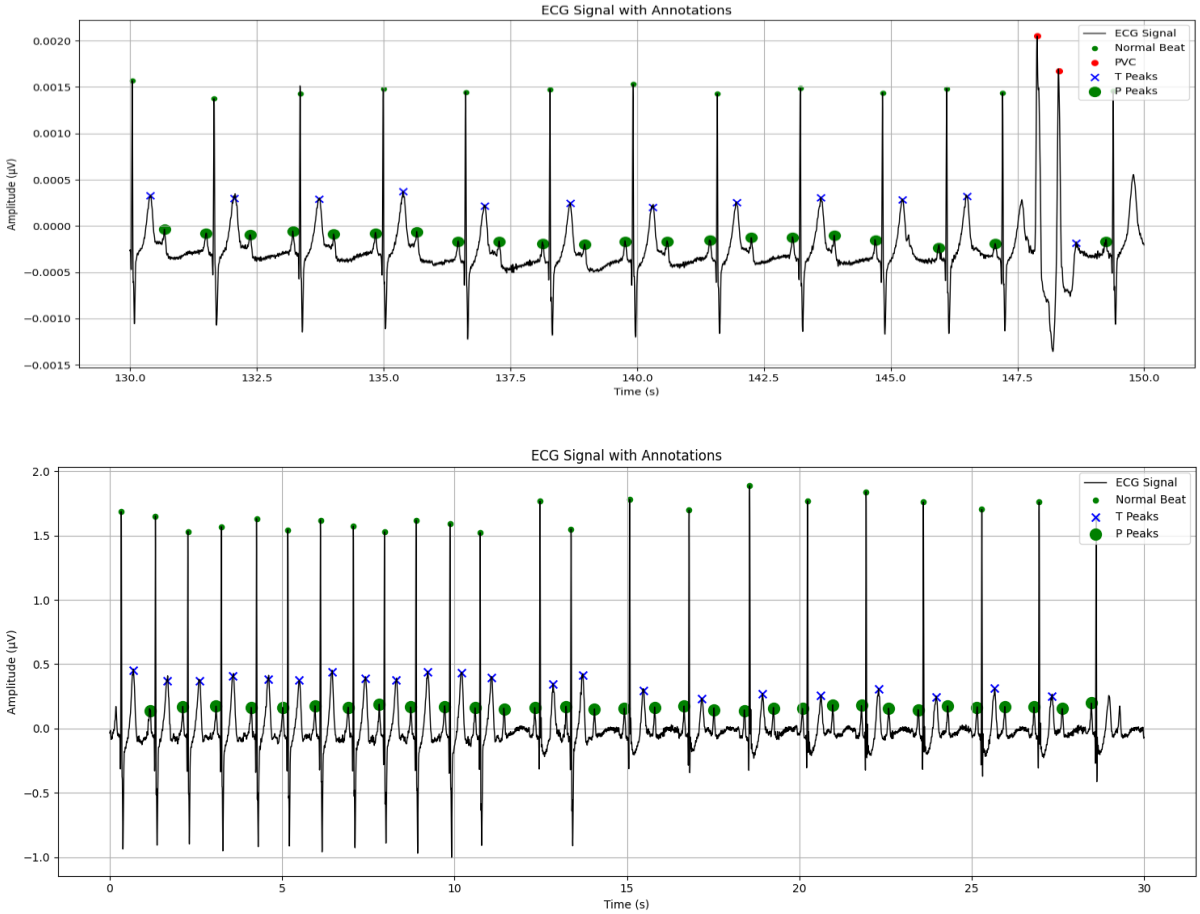
Figure 03: effect of phasor transform

Step 9: Map Peaks to Full ECG

Function: map_peaks_to_real_segments()
Detected peaks are mapped back from the template domain to the full-length ECG signal. Then a local max search happens to detect the max of the relevant window of +_10 samples.

Step 10: Dissociated P-Wave Detection

Function: predict_dis_p_waves_int()
To identify dissociated P-wave (those occurring independently of the normal sinus rhythm) an adaptive rule-based method was implemented using RR interval variations and premature ventricular contraction (PVC) labels.



Example of dissociated P waves

Let rr_intervals[i] represent the RR interval between R-peak i and i+1. The algorithm maintains a boolean list p_dis[] that stores whether each beat is potentially dissociated. The decision for each beat depends on the dissociation status of the previous beat (p_dis[i - 1])

a. If the previous beat was dissociated:
- The current beat is considered dissociated if $RR[i] > 0.8 \times RR[i-1]$

b. If the previous beat was *not* dissociated:

- Stricter conditions are applied: RR[i]>1.6×RR[i−1] and RR[i]>1.6 seconds and not a PVC

Then generate Dissociated Templates and find peaks as before

Function: p_dis_temp_gen()
Using the flagged RR intervals, new templates are generated and peaks are detected between the end of T and the start of the P regions.

- Crop templates between the detected T-wave and the estimated P-wave location
- Apply the same template clustering and peak detection logic

Step 11: Final Algorithm

Function: algorithm ()
All detected peaks (standard and dissociated) are combined and returned. Optional plotting steps allow for visual review.

- Output: det_p (P-wave peaks), det_t (T-wave peaks

Overall wrap

Function: detect_p()
Wraps the entire pipeline. This loads the signal files as .h5py and load annotations from WFDB form for P-peaks for relevant datasets. We can specify the dataset, file, start-end time, plotting etc. Outputs: Dictionary of detected P-wave indices, performance metrics with keys of record names.

Results and Discussion

This method was evaluated using four main datasets from Physio net. Their details are as follows:

MIT-BIH P-wave Dataset:
This dataset contains manually annotated R-peaks and specialist-reviewed P-wave annotations from The free software tool SignalPlant for 12 signals, each 30 minutes long, selected from the MIT Arrhythmia dataset. It includes 5 pure physiological signals without pathologies, and 7 signals with pathological conditions such as AV blocks, AFIB, AFL, pause, and PVCs. These 12 signals cover more than 22000 of total annotated beats.

LUDB (Lviv University Database):
The LUDB contains ECG signals recorded from 40 subjects with diverse pathologies, including both healthy and diseased individuals. It includes 200 of 10 sec, 12-lead ECG recordings with annotations for P-waves, QRS complexes, and T-waves. Here only the lead II was used. The dataset is commonly

used for evaluating algorithms for automatic ECG classification, particularly in detecting arrhythmias and other abnormalities. It contains a variety of pathologies such as AFIB, PVC, and other rhythm disorders. These 200 signals cover more than 1400 of total annotated beats.

QTDB (QT Database):
The QTDB is a large collection of annotated ECG signals primarily used for QT interval measurement and arrhythmia detection. It consists of 105 ECG recordings, including both healthy and pathological signals, sourced from a range of databases. The dataset is known for its focus on arrhythmia detection, particularly in the context of QT interval prolongation, and includes several pathologies such as AFIB, PVC, and other types of ventricular and supraventricular arrhythmias. The QTDB provides manually annotated R-peaks and P-waves, for at least 30 beats for 75 signals and here q1c annotation was used in evaluation. These signals cover more than 2500 of total annotated beats.

BUT PDB (Bologna University Database):
The BUT PDB is a new collection of ECG signals with comprehensive annotations for P-waves, R-peaks, and T-waves. This is consist of 50 signals of 2min length. It includes both healthy and pathological signals, with a particular focus on arrhythmia such as AFIB, PVC, and AV blocks. The database consists of signals from various sources, including patients with different cardiovascular conditions. These signals cover more than 5000 of total annotated beats.

Dataset	Method	PPV (%)	Sensitivity (%)
MIT-BIH (without 207)	This work	96.1	95.4
	Reliable P	96.47	97.2
	Improved Pan-Tompkins	—	94.78
But PDB	This work	89.6	89.1
	Reliable P	88.6	93.07
	Advanced P	79.67	78.3
LUDB*	This work	90.6	97.5
QTDB	This work	90.3	93.2
	EKF	98.6	98.04
	WT	91.07	98.04
	Reliable P	99.1	99.2
	Template-based	96.08	96.05
	Dynamic-based	95.39	96.4
	Differentiation-based	94.73 (99.5**)	95.8 (99.3**)
	GBI-based	94.3	95.67

*This database has mostly been used for onset-offset detection tasks using a 12-lead approach.
**here two papers had mentioned two values.

Upon reviewing the four datasets, it is obvious that some false positives should be addressed. However, when the rule set is adjusted based on one dataset, a reduction in performance across other datasets is often observed. In many cases of FPs, it is noted that AFLs have a considerable effect. Additionally, mis annotations from the databases may contribute to these issues sometimes. Furthermore, during the peak selection process, peaks that are very close to the P-peak are sometimes identified, which can be misleading, even though they fall outside the ±75ms window. Some studies have applied a ±160ms window, which could potentially improve performance. If this approach is adopted, it is anticipated that the performance values could exceed the current results.

Based on these observations, it is concluded that while peak detection is generally effective, the exact selection of the peak may sometimes be misleading. Therefore, the inclusion of an AFL detection algorithm, improvements to peak selection logic, and refinement of R-peak-based peak rejection criteria are expected to reduce the occurrence of false positives in this method.

Regarding the sensitivity values for LUDB and the five physiological signals from MIT, the results are considered to be at the state-of-the-art level. However, when QTDB is considered, it is found that there are methods demonstrating better performance. Limitations in performance evaluation criteria of various methods need to be taken into account. For instance, certain methods use only part of the QTDB database, while some 12-lead-based approaches for P-wave detection use all the leads and select the highest-valued lead for each signal. This method can be advantageous in scenarios like AFL, where some leads may not be affected. Furthermore, the results obtained from pathological databases like BUT PDB are promising, as they are comparable to the performance of other methods. Here let’s discuss about the possible FPs of this algorithm.

This works well in most signals, but in MIT records 207 and 222, there are some considerations for the algorithm.

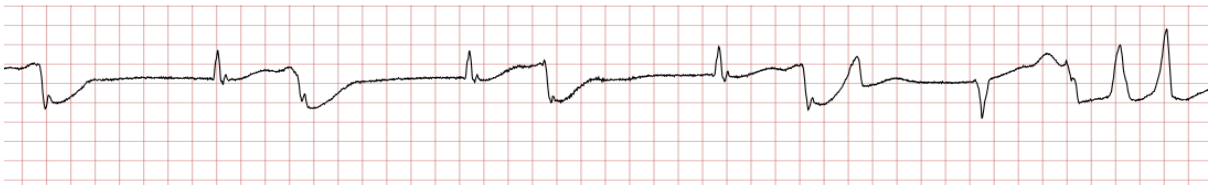


Figure 05: 207 signal from MITBIH

When signal number 207 is considered, it is very difficult to determine whether a P-peak is present. Furthermore in some scenarios researches have excluded that 207 signal from the evaluation process considering that it can contain misannotations. And in signal 222 from MITBIH database is highly affected with AFL signal part which leads to higher number of FPs.

In some instances, false postives for AFIB and PVC detection can affect the P-wave detection process, as these beats are entirely excluded from the P-wave analysis. However, the majority of the databases used contain very short signals, which can impact the performance of the AFIB detection algorithm, particularly as many of these signals contain fewer than 128 beats. Therefore, it is necessary to test this method with longer signals before drawing any conclusions. The only available source for manual P-wave annotations for longer signals is the MIT-BIH P-wave database. While this database is comprehensive and includes many pathologies, it contains only 1.5 minutes of AFIB data, which is likely insufficient for thoroughly addressing this issue.

Ultimately, the phasor transform method appears to be a promising option for performing P-wave detection on ECG signals in edge devices. This method has been supported by literature for an extended period. However, many existing methods lack a reliable approach for pathological scenarios, which is why this particular aspect has been emphasized.

The paper referenced has served as the primary resource for this process. It is suggested that by combining AFIB and PVC detection, along with statistical-based AFL detection, false positives in peak detection could be minimized through the use of the phasor transform, making it a highly promising method for P-peak detection on edge devices.

The main concern with this method is the potential for the misidentification of a peak within the template, which could affect the entire segment set. However, due to the strict limitations of the window definitions based on heuristic rules, the likelihood of this occurring is low. For future work, this method provides a strong foundation for a machine-learning-based approach to extract valuable features from the signal. The next iteration of this method could involve the integration of AFIB, PVC, and AFL detection with the template and phasor-based feature extraction method, which would define interpretable model of physiological rules to select the best peak from the templates.

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